

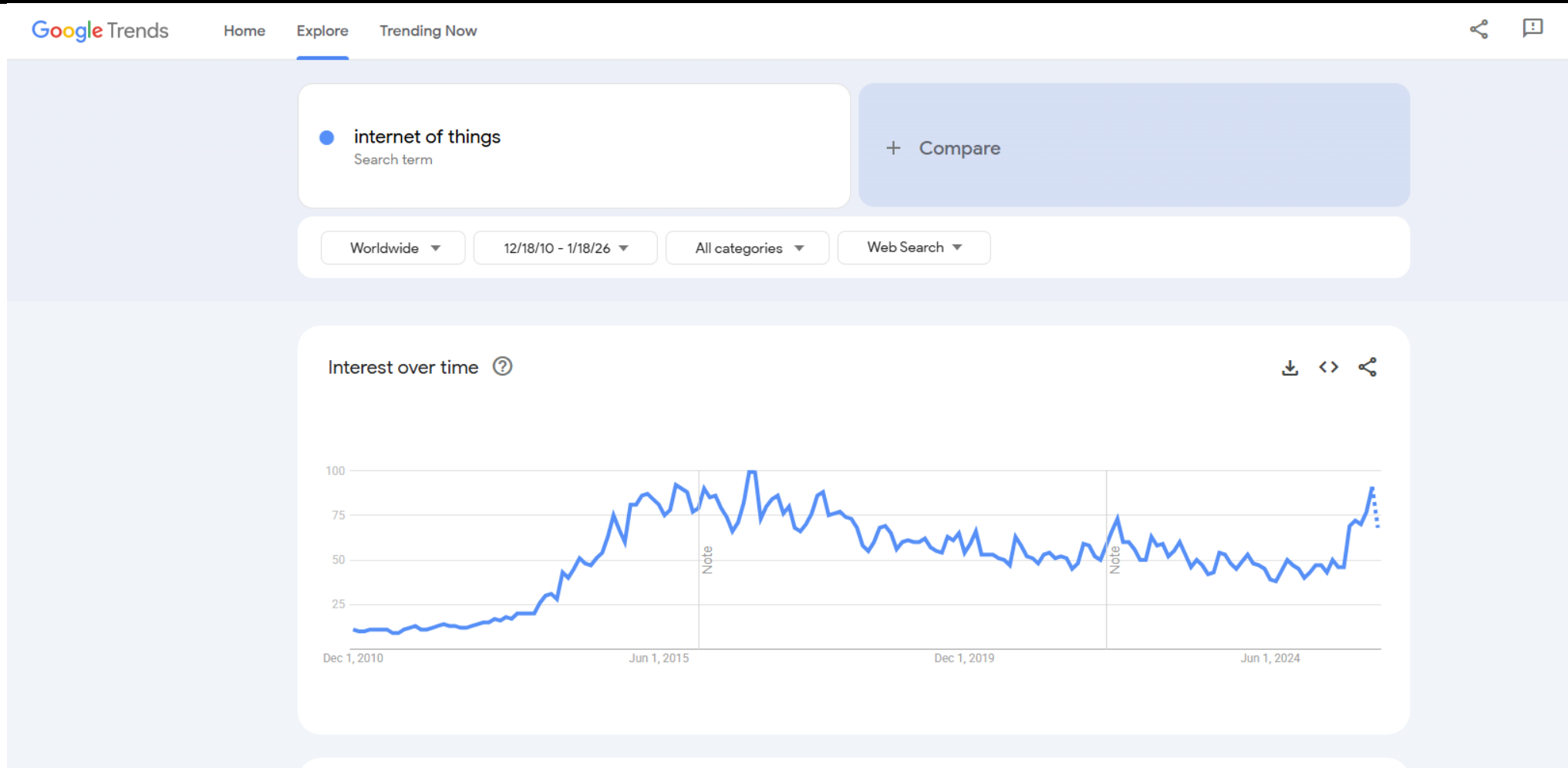
Internet of Things (IoT)

Concepts and applications

Agenda

- What is IoT?
- From idea to reality
- IoT applications
- IoT system layers

To The Internet of Things (Google trends)



To The Internet of Things (academics)



Suggested publications:

[View all](#)



210,608 results

[sorted by relevance](#) | [date](#)

Refine by:

Years

- ☐ 2026 (5,666)
- ☐ 2025 (24,935)
- ☐ 2024 (22,059)
- ☐ 2023 (18,816)
- ☐ 2022 (18,565)
- ☐ 2021 (16,533)
- ☐ 2020 (13,867)
- ☐ 2019 (11,111)

Research article

Implementing TinyML in Internet of Things Devices: A Systematic Literature Review

Journal of Industrial Information Integration, Available online 9 January 2026

Andrés Felipe Solís Pino, Daniel Steven Moran Pizarro, ... Fernando Moreira

Research article

Reliability-centered approach to artificial intelligence-driven predictive maintenance for industrial

Internet of Things

Engineering Applications of Artificial Intelligence, 1 January 2026

Wei Wu

Suggested topics [?](#)

Internet of Things

in Computer Science

Internet of Things

in Engineering

Internet of Things

in Psychology

What is the IoT?

1. **A buzzword**
2. **A global network of connected objects**
3. **A technology**
4. **A concept**
5. **A new type of communication network**
6. **Another way of talking about AI**
7. **Not important**

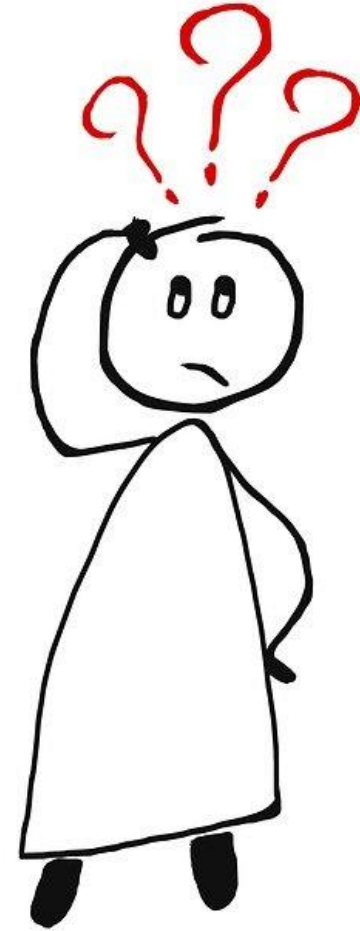


Image par [ElisaRiva](#) de [Pixabay](#)

Origin of IoT



1999!

**"THE INTERNET OF THINGS IS
ABOUT EMPOWERING COMPUTERS
...SO THEY CAN SEE, HEAR
AND SMELL THE WORLD FOR
THEMSELVES"**

**KEVIN ASHTON
INVENTOR OF THE TERM
"INTERNET OF THINGS"**



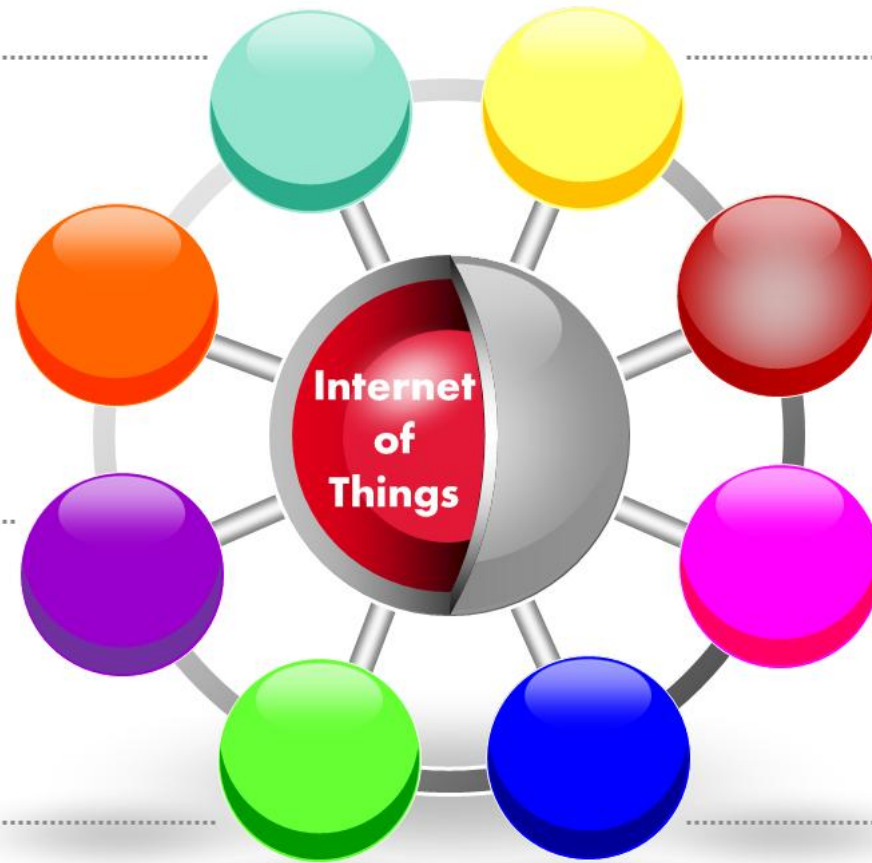
IoT –The vision

A dynamic global network infrastructure

with self configuring capabilities

based on standard and interoperable communication protocols

where physical and virtual “things”



have identities, physical attributes, and virtual personalities

use intelligent interfaces,

and are seamlessly integrated

into the information network.

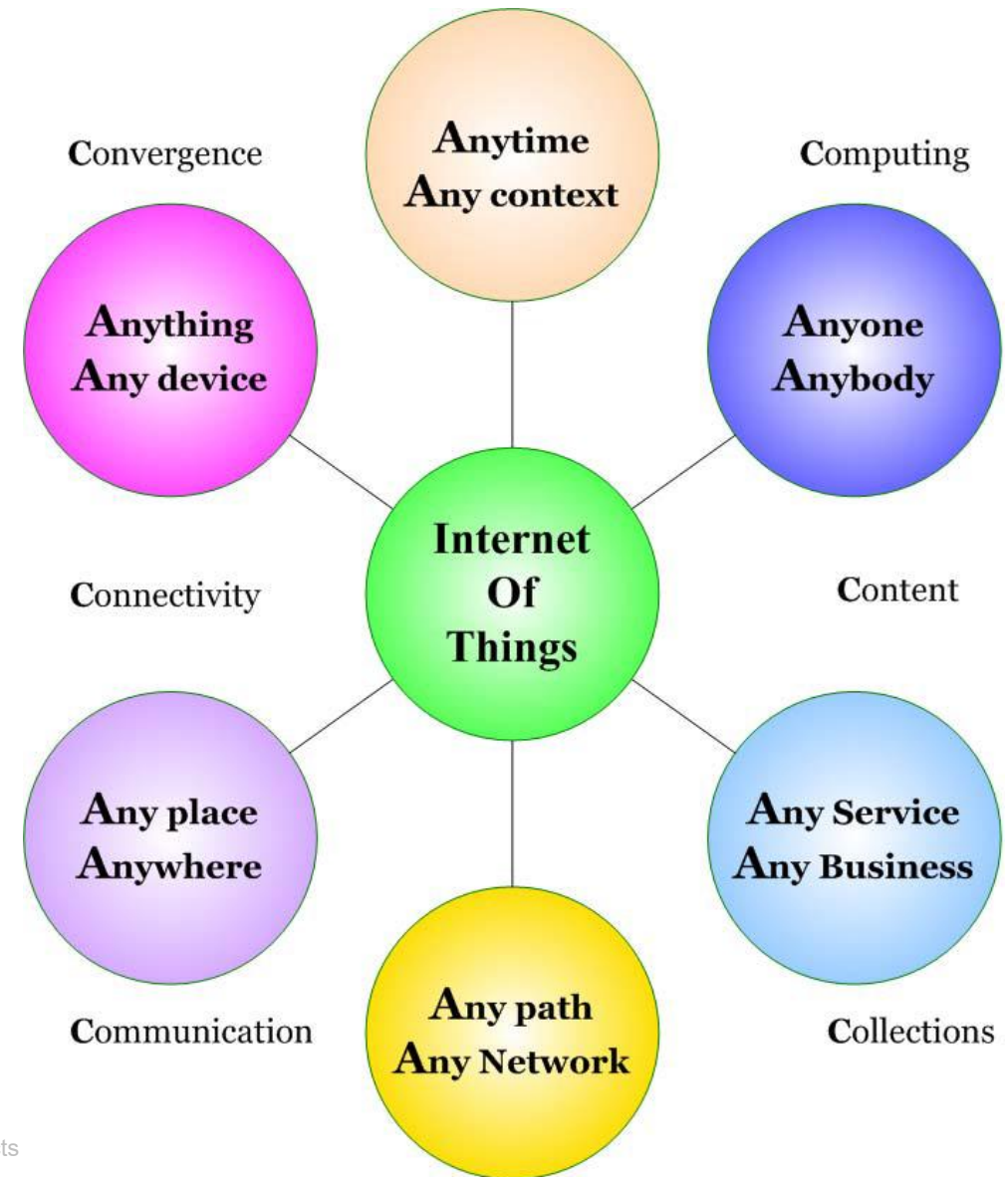
http://www.internet-of-things-research.eu/about_iot.htm

The IoT

The idea

Allows people and things to be connected

- Anytime,
- Anyplace, with
- Anything and
- Anyone, ideally using
- Any path/network and
- Any service



CERP-IoT, (2009) "Internet of Things: Strategic Research Roadmap", Report from the Cluster of European Research Projects
Internet of Things (CERP-IoT). Available at: http://www.internet-of-things-research.eu/pdf/IoT_Cluster_Strategic_Research_Agenda_2009.pdf

Opening **new business models**

The Internet of Things

- **Concept** in which each **unique** objects (living or not) is equipped with **a device** enabling it to **communicate & connect automatically** in **real time** with its environment (physical & virtual) and to manage its owns **transactions**

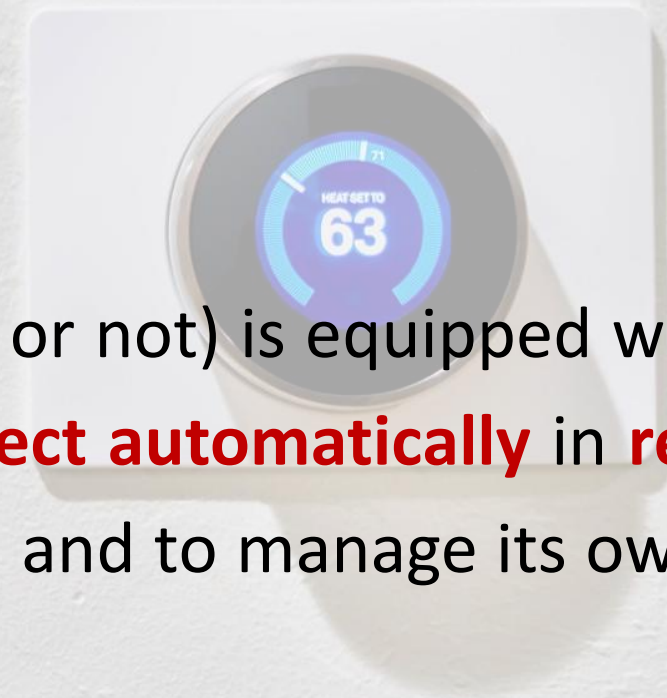
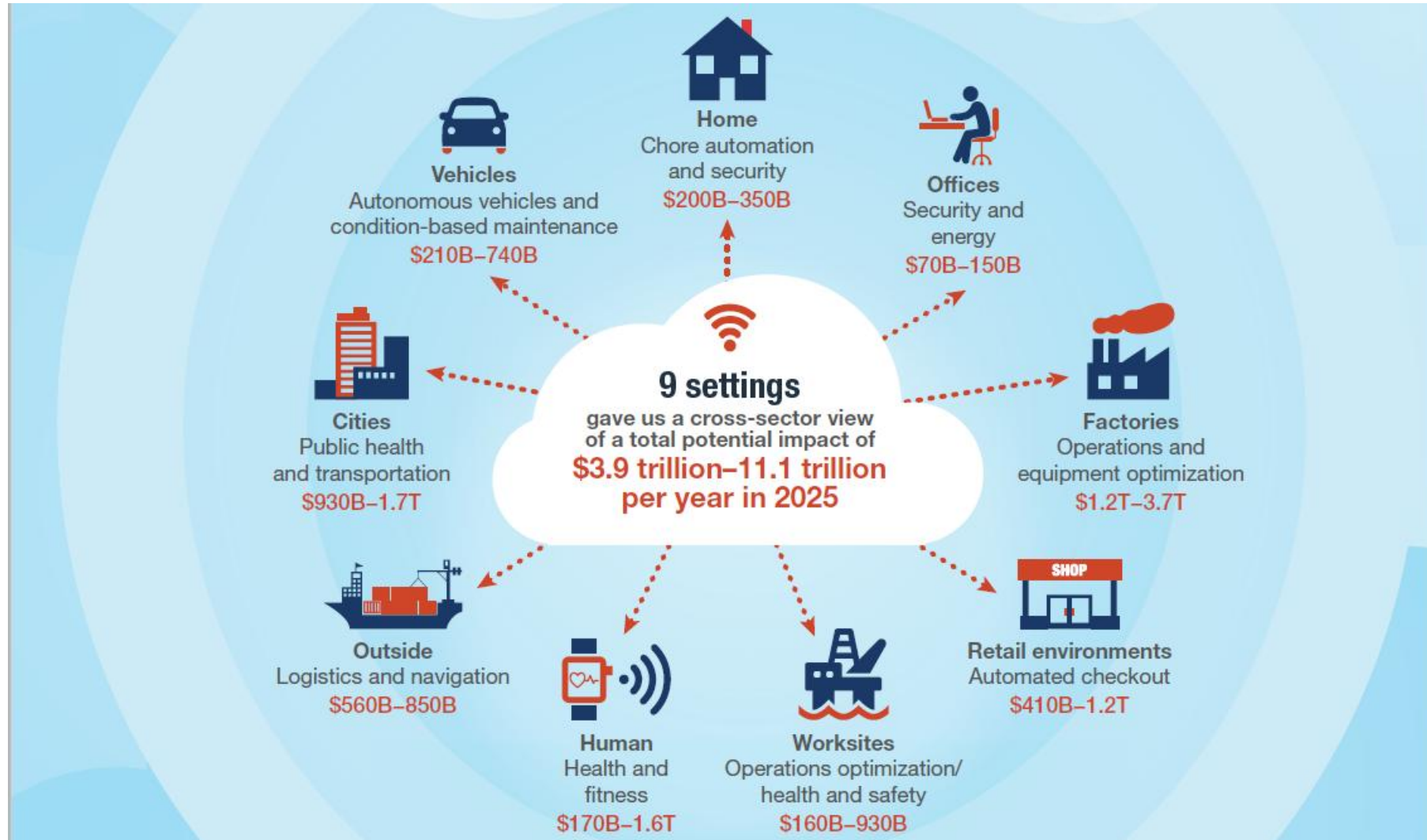


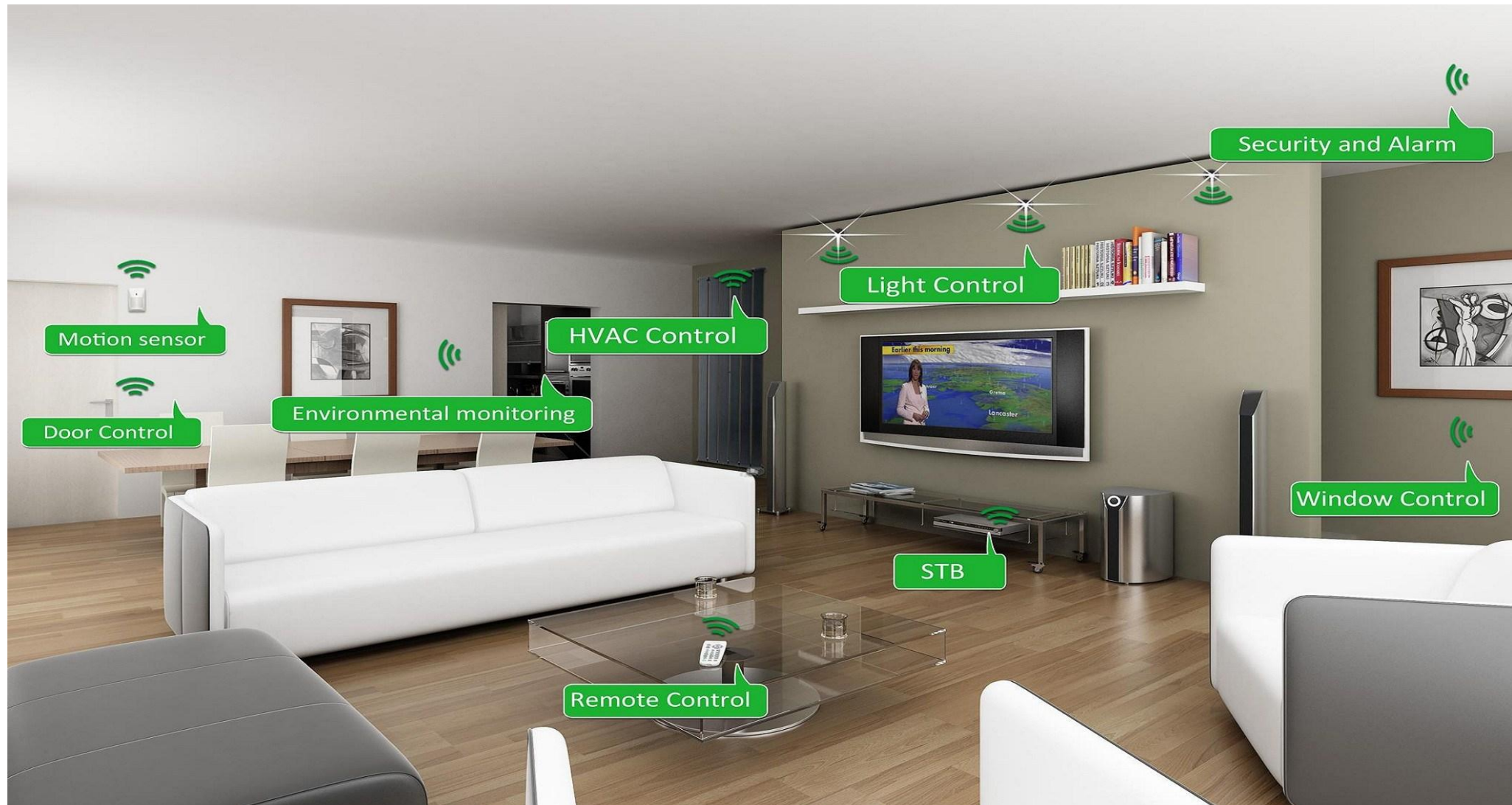
Photo by Dan LeFebvre on Unsplash-light

Where is the value potential of the IoT?



Source: McKinsey & Cie, THE INTERNET OF THINGS MAPPING THE VALUE BEYOND THE HYPE, 2015

What is the (every)Thing?



What is the (every)Thing? (Cont.)

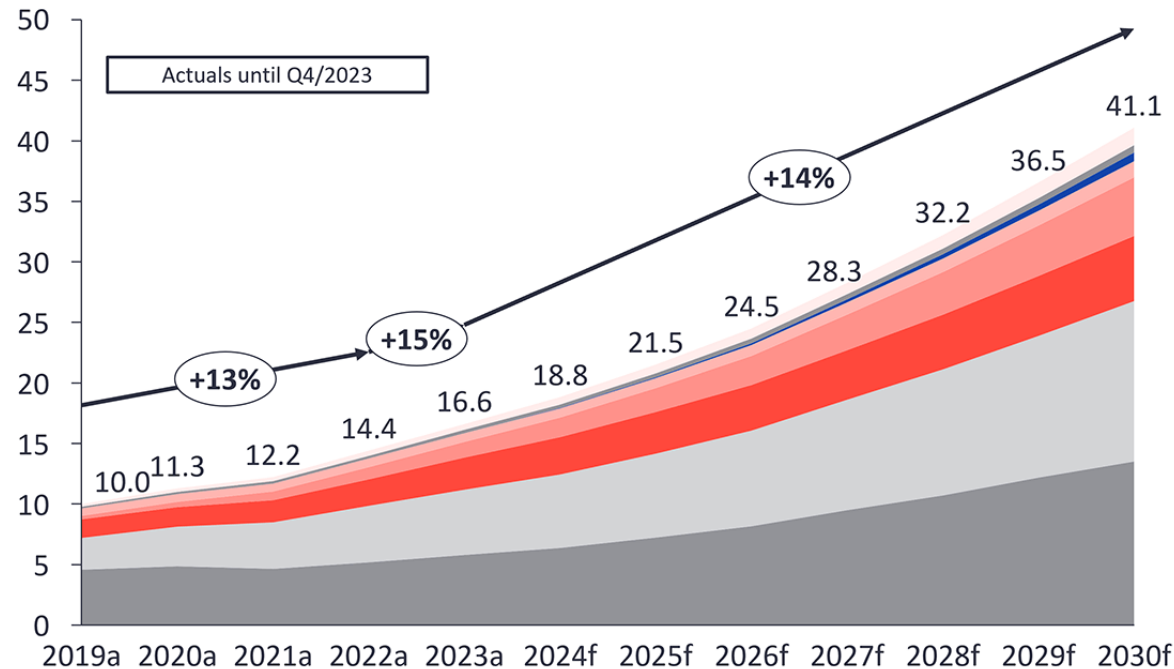


<http://blogs.jgh.ca/jghnews/robotic-surgery-gaining-ground-at-the-jgh/>

Connected IoT Devices Forecast 2024–2030

Global IoT market forecast (in billions of connected IoT devices)

Number of global active IoT connections (installed base) in billions



Connectivity type	CAGR 21–23	CAGR 23–30
Other	21%	17%
Wireless neighborhood area networks (WWAN)	15%	14%
Cellular 5G IoT	147%	62%
Wired IoT	4%	9%
LPWA	35%	21%
Cellular IoT (excl. 5G, LPWA)	21%	11%
Wireless local area networks (WLAN)	18%	14%
Wireless personal area networks (WPAN)	12%	13%

XX% = CAGR

Note: IoT connections do not include any computers, laptops, fixed phones, cellphones, or consumers tablets. Counted are active nodes/devices or gateways that concentrate the end-sensors, not every sensor/actuator. Simple one-directional communications technology not considered (e.g., RFID, NFC). Wired includes ethernet and fieldbuses (e.g., connected industrial PLCs or I/O modules); Cellular includes 2G, 3G, 4G, 5G; LPWA includes unlicensed and licensed low-power networks; WPAN includes Bluetooth, Zigbee, Z-Wave or similar; WLAN includes Wi-Fi and related protocols; WWAN includes non-short-range mesh, such as Wi-SUN; Other includes satellite and unclassified proprietary networks with any range.

Source: IoT Analytics Research 2024-State of IoT Summer 2024. We welcome resharing: Please attribute this image to its original source and include a link back to the original article.

To everywhere...

Les objets connectés font leur entrée dans la salle de bain



<https://ici.radio-canada.ca/nouvelle/1467879/objets-connectes-intelligents-salle-de-bain-connectee-intelligente>

Park, S., Won, D.D., Lee, B.J. et al. A mountable toilet system for personalized health monitoring via the analysis of excreta. Nat Biomed Eng (2020). <https://doi.org/10.1038/s41551-020-0534-9><https://www.nature.com/articles/s41551-020-0534-9>

A mountable toilet system for personalized health monitoring via the analysis of excreta



To the reality...

Application in the retail sector



Introducing Amazon Go and the world's most advanced shopping technology

<https://www.youtube.com/watch?v=NrmMk1Myrxc>

To the reality...

Application in the retail sector – to Amazon Fresh



We're excited to launch the all-new Amazon Dash Cart, which helps make a quick shopping trip even quicker by skipping the checkout line. Jeff Helbling, vice president, Amazon Fresh Stores, August 27, 2020

<https://www.aboutamazon.com/news/retail/introducing-the-first-amazon-fresh-grocery-store>

www.amazon.com/dashcart

<https://www.youtube.com/watch?v=tRmPGqnzn7o>

IRVINE

Amazon's Newest & Biggest Grocery Store: Amazon Fresh Irvine

Up to 2.45 min

To the reality...

Application in the retail sector & how it works



We go inside Amazon's brand-new grocery store (no cashiers in sight)

<https://www.youtube.com/watch?v=ITzPpAbjasA>

But this not so new!



IBM Commercial - RFID

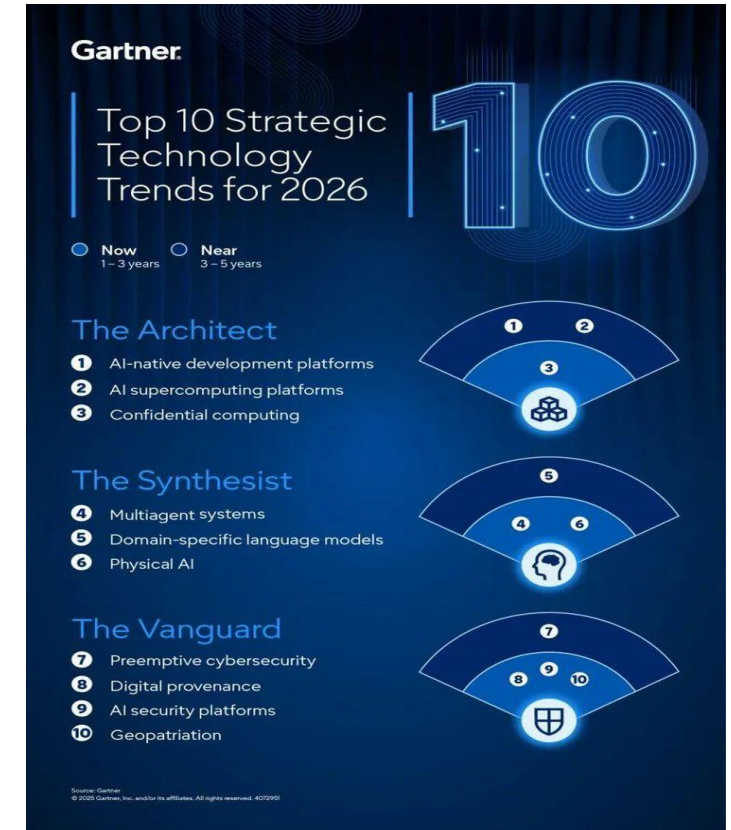
<https://www.youtube.com/watch?v=oAvQcYcvyaw>



IBM RFID Commercial-The Future Supermarket

<https://www.youtube.com/watch?v=wzFhBGKU6HA>

Technological trends



“As organizations improve not only the amount of data they capture but also how they combine data from different sources and use that data, the IoB (Internet of Behavior) will continue to affect how organizations interact with people ”

Brian Burke, Research Vice President, during virtual [Gartner IT Symposium/Xpo™ 2020](#).

<https://www.gartner.com/smarterwithgartner/gartner-top-strategic-technology-trends-for-2021/>

THE INTERNET OF EVERYTHING

I TOLD YOU TO PICK UP MILK ON THE WAY HOME.
DON'T YOU LISTEN TO
ANYTHING I SAY?!!

Don't forget
to buy milk!

DING

Cartoon for CloudTweaks.com

<http://cloudtweaks.com/humor/>

From the Concept...to the reality



<https://www.tersosolutions.com/solutions/rfid-cabinets/>

Any other application based on the same Concept

1. In retail
2. In health care
3. In restaurants
4. In manufacturing
5. In smart cities
6. In novative 😊

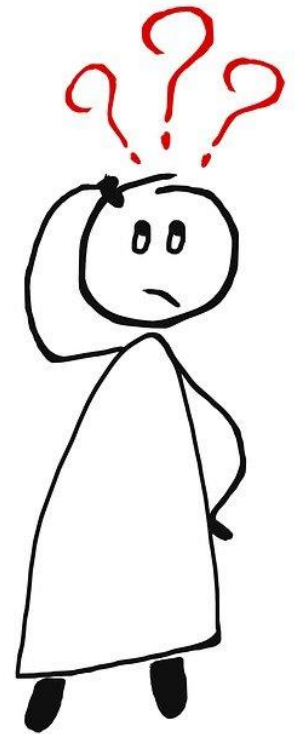
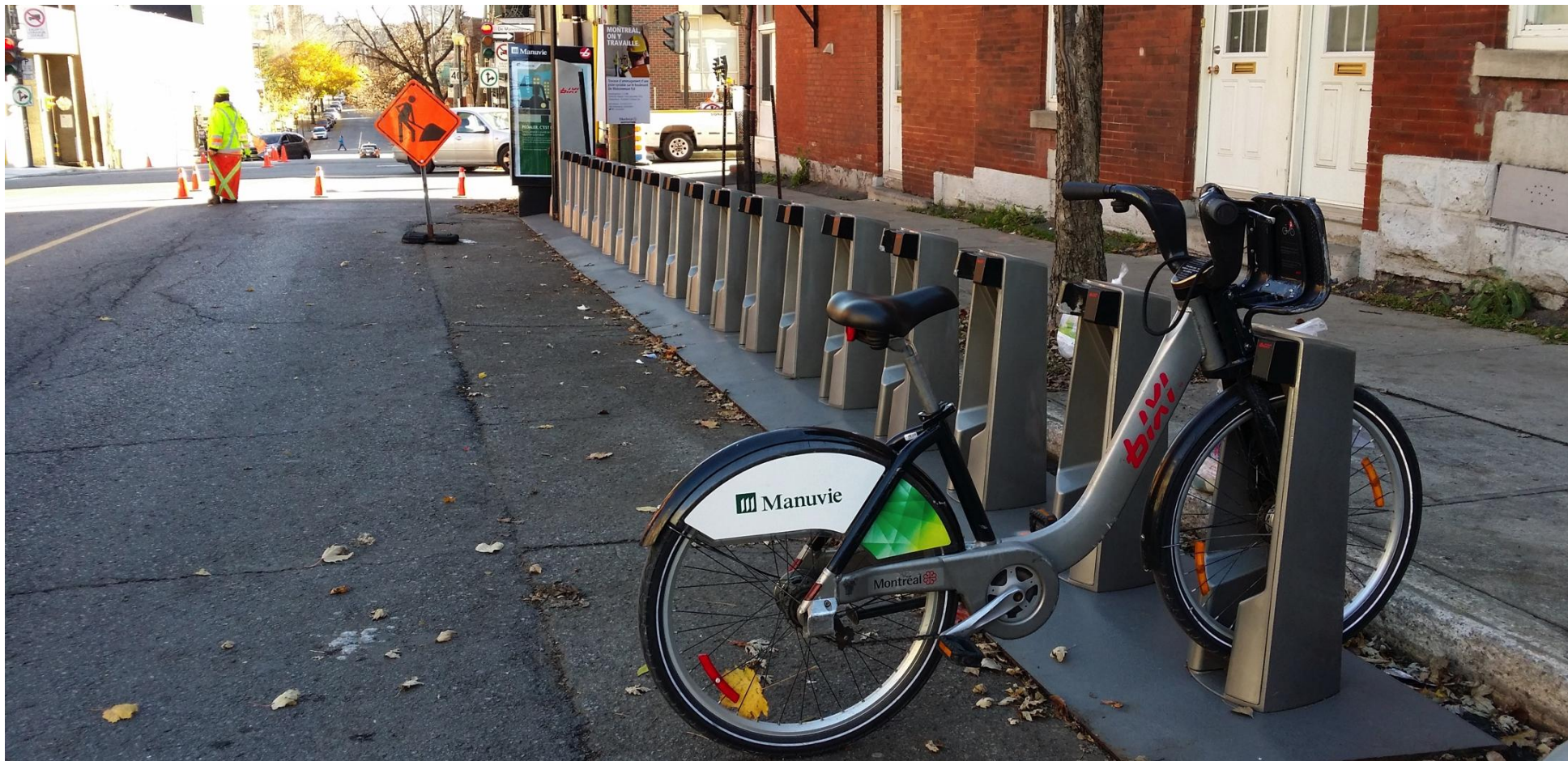


Image par [ElisaRiva](#) de [Pixabay](#)

Same concept– different context

Bixis



Same concept— different context

Martini Smart Cube

Martini's Smart Cube, detects when a glass is empty and automatically triggers a reorder of that drink, while also enabling bar personnel to quickly locate an individual whose drink requires replacement



<http://www.rfidjournal.com/articles/view?15044>

Same concept— different context

Suchis automated replenishment



Source: <https://www.pre-motion.nl/lopende-band-food-serveren/sushiband-restaurant>

Get connected....



<https://www.c2montreal.com/fr/news-fr/oubliez-votre-portefeuille-et-laissez-votre-badge-connecte-faire-tout-le-travail/#/>

IoT - a metaphor

Behind the scene



Idée reprise de IdO Montreal 2015 – conf. De Carlo Rati - <http://senseable.mit.edu/>

So how would you represent the IoT

1. Technological perspective
2. Business perspective
3. Both

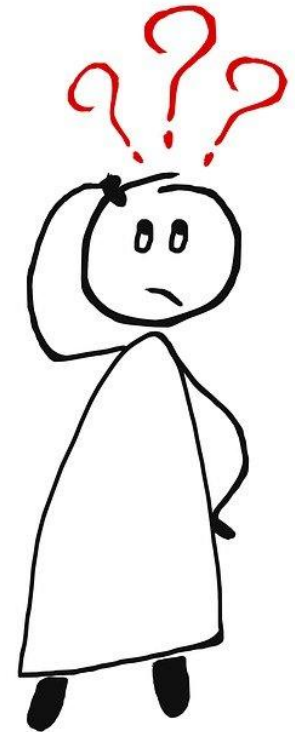
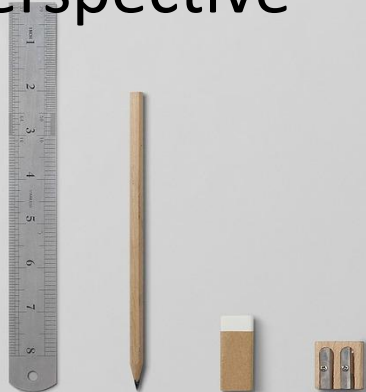
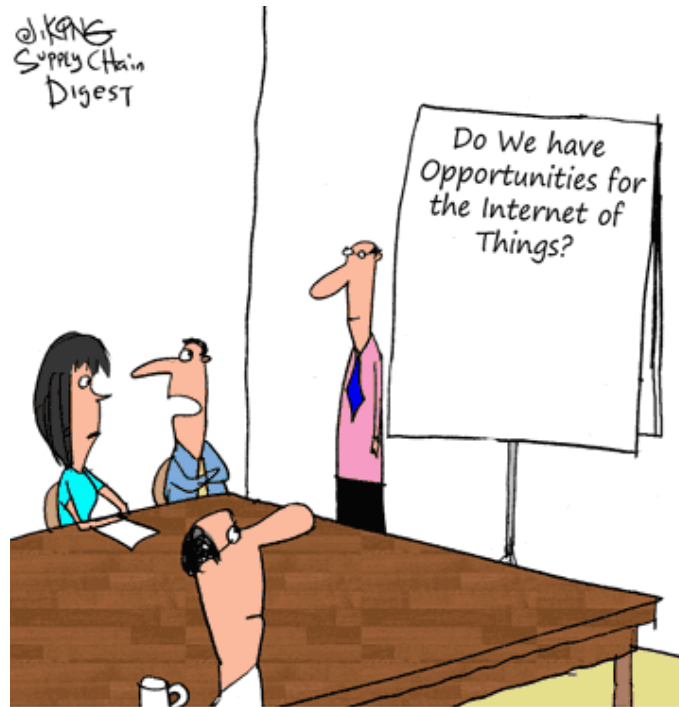
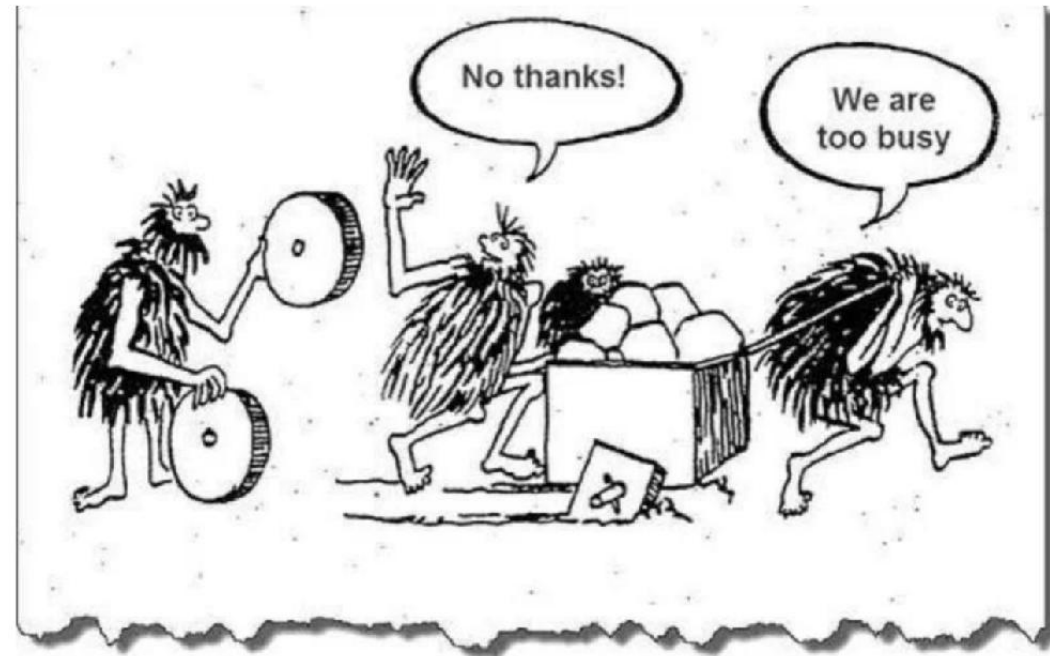


Image par [ElisaRiva](#) de [Pixabay](#)

Ready for the IOT?



"We have things. We have the internet. That's a start."



Layers of an IoT system

Multi technos/multi standards/multi protocols

